
Adaptive-Mixture Augmented Policy Gradients for Expected-Length-Constrained Reinforcement Learning with Verifiable Rewards

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Abstract

Reinforcement learning with verifiable rewards spends most of its compute on fresh language-model trajectories, while ordinary replay is statistically problematic because sequence likelihood ratios become unstable and previously consumed samples depend on the current policy. We study a narrower alternative: at each frozen policy, combine fresh on-policy trajectories with newly generated, single-use trajectories from a known stale actor and an adaptive prompt distribution. A current-policy mixture component bounds the exact joint importance ratio, and frozen reward and length models supply an augmented control variate. Under explicit sampling and decoding assumptions, the resulting one-update gradient estimator is conditionally unbiased. A Lagrange multiplier constrains expected response length instead of silently normalizing the objective per token. This methods-first draft gives the estimator audit and matched-budget training protocol, but reports no performance gain before those experiments are run.

1. Introduction

Language-model reinforcement learning with verifiable rewards (RLVR) can train reasoning behavior from executable tests or exact answer checks. Its simplicity does not remove the cost of generation: a policy update may require many long trajectories, most of which are discarded after one use. Critic-free descendants of REINFORCE (Williams, 1992), such as RLOO (Ahmadian et al., 2024) and GRPO (Shao et al., 2024) reduce value-model overhead, while large-scale systems add clipping, prompt filtering, and token-level aggregation (Yu et al., 2025). These choices improve engineering behavior but change which prompt, rollout, or token

population receives weight.

Reusing trajectories appears attractive, yet “off-policy” covers distinct statistical settings. A replay-buffer trajectory may have selected itself into the buffer through its reward and may already have changed the current policy. Recomputing its likelihood does not restore independence. Sequence-level importance ratios can have extreme variance; token-level ratios do not represent the sampling probability of a sequence-rewarded trajectory. Practical off-policy methods therefore use tapering, buffers, or adaptive clipping (Le Roux et al., 2025; Wan et al., 2026; Xi et al., 2026), knowingly trading bias for stability.

We ask whether a controlled source of stale data can improve gradient efficiency without making an inaccurate replay-unbiasedness claim. At iteration t , a live stale actor produces *new*, never-consumed trajectories from a logged policy and prompt distribution. These are combined with a positive fraction of current-policy trajectories. An augmented importance estimator uses a frozen outcome predictor to reduce residual variance, then takes exactly one policy update and consumes the data. This does not reuse rollout FLOPs: its plausible benefit is lower residual-gradient variance per total generation and scoring cost when the alternative proposal reaches informative regions. Current and stale actors may generate in parallel while the learner remains frozen. Learner-actor asynchrony is a separate non-theorem practical variant, not a claimed benefit of the core algorithm.

Our contributions are deliberately scoped:

- We define the target joint trajectory law, including the prompt and decoding distributions, and derive a stratified adaptive-mixture augmented policy-gradient estimator with bounded exact density ratio.
- We state sufficient conditions for conditional unbiasedness and the precise two-route bias identity. The result excludes ordinary replay, multiple epochs, clipping, and group standardization.
- We couple the estimator to an explicit expected-token constraint and provide logging and diagnostics that can falsify its assumptions.

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- We specify a frozen-policy estimator audit and a one-update-per-batch matched-budget study across model families and verifier domains.

Evidence status. No downstream or efficiency result is asserted in this draft. The theorem concerns one raw gradient estimate at a frozen policy; it neither guarantees a better optimizer step nor policy improvement.

2. Related Work

On-policy and group-relative RL. Careful PPO reproduction shows that reward whitening, KL treatment, clipping, and response length are implementation-sensitive (Huang et al., 2024). RLOO gives a simple leave-one-out REINFORCE baseline (Ahmadian et al., 2024); GRPO standardizes rewards within prompt groups (Shao et al., 2024); Dr. GRPO questions group-standard-deviation and response-length normalization (Liu et al., 2025b); and DAPO adds dynamic sampling, asymmetric clipping, and token-level aggregation (Yu et al., 2025). GSPO uses a length-normalized geometric mean of token ratios with response-level clipping (Zheng et al., 2025). Recent analyses expose negative-gradient pathologies (Deng et al., 2025), U-statistic structure (Zhou et al., 2026), and the interaction between clipping, entropy, and misaligned rewards (Chen et al., 2026). Our estimator intentionally omits group normalization and PPO clipping so its target expectation remains explicit.

Allocation, baselines, and length. Adaptive rollout allocation predicts prompt difficulty and assigns rollout counts under a fixed budget (Nguyen et al., 2026). Cross-prompt shrinkage baselines reduce variance while preserving a policy-gradient target (Zeng et al., 2026). Group-relative reward rescaling addresses length inflation (Li et al., 2026). We instead represent length as a constrained cost and adapt the stale-source prompt proposal; these components are not claimed to subsume those methods. Discounted Beta-Bernoulli reward estimation reuses historical reward statistics to reduce group-estimator error while explicitly accepting bias (Kim et al., 2026). It is therefore a required bias-variance comparator: reusing summary statistics is distinct from our density-corrected use of newly drawn stale-policy trajectories.

Off-policy stability and reward validity. TOPR treats positive examples with an SFT/log-ratio-like term and truncates importance weighting for negative examples (Le Roux et al., 2025); buffer-based and balanced policy optimization improve practical reuse (Wan et al., 2026; Xi et al., 2026). Such stabilized estimators may be preferable even when biased. Separately, spurious-reward controls show that benchmark gains need not demonstrate learning from the intended verifier (Shao et al., 2026). The protocol includes those

controls and treats exact statistical assumptions, not empirical stability alone, as the object of the first study. Direct replay studies optimize the generation–staleness–diversity trade-off (Arnal et al., 2026), while POPO prioritizes effective recent groups and uses decoupled off-policy correction (Mao et al., 2026). Both are stronger practical comparators than naive reuse. Their empirical utility does not imply the conditional unbiasedness claimed for our newly sampled, single-use design. Asynchronous RLHF separates generation from learning on lagged policies (Noukhovitch et al., 2025); it motivates the stale-actor systems layout but does not provide our conditional estimator identity. Statistically, the design is a restricted stratified multiple-importance sampler with augmentation. Classical work optimizes mixture allocation and control variates jointly (He & Owen, 2014); our pilot deliberately studies a small fixed grid rather than claiming a new general MIS principle or optimal allocation.

PrefixRL and PROS instead reuse an old prefix while generating a new continuation (Setlur et al., 2026; Huang & Wan, 2026); they change the prompt or conditional objective and avoid full-trajectory ratios. They are practical compute comparators, not instances of Equation (8). At the systems layer, HybridFlow decouples RL control and distributed computation (Sheng et al., 2025); our implementation follows that separation so rollout, cross-policy scoring, and update costs remain individually measurable. In its exact-acceptance setting, SPEC-RL uses old trajectories as speculative drafts while preserving the current-policy rollout law; its lenient setting does not have that exact claim (Liu et al., 2025a). MinPRO instead replaces exact causal-prefix ratios by a stabilized minimum-ratio surrogate under policy lag (Lei et al., 2026). These methods target complementary systems and biased-stability trade-offs.

3. Formal Setting

Let $Z = (X, Y)$ contain prompt X and a complete generated sequence Y , including EOS or a declared truncation event. The target prompt distribution is D , a versioned finite prompt multiset with known masses $d(x)$, and the exact rollout distribution is $\pi_\theta(y \mid x)$. Thus

$$P_\theta(z) = d(x)\pi_\theta(y \mid x). \quad (1)$$

The rollout decoder is part of π_θ . Temperature, truncation, and any rejection rule must be represented by its normalized sampling probability; raw base-model logits are not automatically the rollout density. The theorem-facing implementation uses temperature-softmax sampling without top- k , top- p , repetition penalties, minimum-length masks, or forced tokens, with fixed maximum length and EOS included in the outcome. The allowed vocabulary $\mathcal{V}_\ell$ is a deterministic, parameter-independent function of the tokenizer and prefix. For temperature T_d the logged per-token

mass is

$$p_\theta(y_\ell \mid y_{<\ell}, x) = \frac{\mathbf{1}\{y_\ell \in \mathcal{V}_\ell\} \exp(z_{\theta,\ell,y_\ell}/T_d)}{\sum_{v \in \mathcal{V}_\ell} \exp(z_{\theta,\ell,v}/T_d)}. \quad (2)$$

The sequence mass is the product through EOS; no-EOS sequences at the fixed limit carry their product mass and a deterministic truncation status. An empirical conformance test for both current and stale sampler–scorer pairs must pass before estimator audits.

The verifier deterministically maps every complete Z , including parse, timeout, sandbox-error, and truncation statuses, to bounded reward $R(Z)$; no outcome is retried or dropped. Interactive or stochastic environments are outside the core formulation. Generation incurs $C(Y) = |Y|$ response tokens. For fixed expected-response budget κ , we optimize the Lagrangian

$$\mathcal{L}(\theta, \lambda) = \mathbb{E}_{P_\theta}[R] - \lambda\{\mathbb{E}_{P_\theta}[C] - \kappa\}, \quad \lambda \geq 0. \quad (3)$$

The constrained problem is $\max_\theta \min_{\lambda \geq 0} \mathcal{L}(\theta, \lambda)$ with $\mathbb{E}_{P_\theta}[C] \leq \kappa$. The stochastic dual update below is a constraint controller, not a finite-sample feasibility guarantee. Define $U_\lambda = R - \lambda C$ and the raw sequence score

$$\begin{aligned} s_\theta(Z) &= \nabla_\vartheta \log \pi_\vartheta^{\text{decoder}}(Y \mid X)|_{\vartheta=\theta} \\ &= \sum_{\ell=1}^{|Y|} \nabla_\vartheta \log \pi_\vartheta^{\text{decoder}}(Y_\ell \mid X, Y_{<\ell})|_{\vartheta=\theta}. \end{aligned} \quad (4)$$

We do not divide Equation (4) by realized length. Under standard score-function conditions, $\nabla_\theta \mathcal{L} = \mathbb{E}_{P_\theta}[U_\lambda s_\theta]$.

4. Method

Condition on the pre-update history $\mathcal{F}_t$. The current policy θ_t , dual variable, nuisance models, and all sampling distributions are frozen before trajectories are drawn. A logged adaptive prompt distribution Q_t and a stale behavior policy μ_t define

$$G_t(z) = q_t(x)\mu_t(y \mid x). \quad (5)$$

G_t is generated only after all iteration- t choices are frozen. A queue whose realized contents can affect θ_t , q_t , α , or nuisance models is not conditionally fresh and does not satisfy the theorem, even if its trajectories have never been optimized on.

For fixed $\alpha \in (0, 1]$, define the joint mixture

$$M_t(z) = \alpha P_{\theta_t}(z) + (1 - \alpha)G_t(z) \quad (6)$$

and exact detached weight

$$w_t(z) = \frac{P_{\theta_t}(z)}{M_t(z)} \leq \frac{1}{\alpha}. \quad (7)$$

The prompt density is part of Equation (7). A conditional policy ratio is valid only when $Q_t = D$. Defensive mixtures and augmented importance estimators are established statistical tools (Owen & Zhou, 2000; Dudík et al., 2011); our contribution is their restricted, auditable instantiation for fresh stale-actor RLVR, not the general identities themselves.

4.1. Learning Objective

Let $h_t^R(z)$ and $h_t^C(z)$ be frozen action-conditional predictors of reward and length, and set $h_t = h_t^R - \lambda_t h_t^C$. This separation allows the dual variable to change without retraining the two predictors. Let $b_t(x)$ be a frozen prompt-only baseline. In stale-source conditions, fix integers $n_P, n_G \geq 1$ and mixture mass $\alpha \in (0, 1)$ independently before sampling; write $\rho = n_P/(n_P + n_G)$ for the allocation fraction. Draw $Z_i^P \sim P_{\theta_t}$ and $Z_j^G \sim G_t$. For $\alpha < 1$, deterministic stratification gives

$$\begin{aligned} \hat{g}_t &= \frac{1}{n_P} \sum_{i=1}^{n_P} [(h_i - b_i)s_i + \alpha w_i(U_i - h_i)s_i] \\ &\quad + \frac{1 - \alpha}{n_G} \sum_{j=1}^{n_G} w_j(U_j - h_j)s_j. \end{aligned} \quad (8)$$

The first line uses current-policy samples and the second uses stale-source samples. All w, h, b are stop-gradient quantities. The $\alpha = 1$ limit omits the stale-source sum and sets $n_G = 0$; it reduces exactly to ordinary on-policy $(U - b)s$. The outcome predictor is critic-like infrastructure; we do not call the method critic-free. The exploratory pilot restricts $\rho = \alpha$ to keep 64 total trajectories; the identity itself does not.

For conditionally IID trajectories within each of two independent strata define $A = (h - b)s + \alpha w(U - h)s$ under P and $B = w(U - h)s$ under G . Conditional on the frozen history,

$$\text{Cov}(\hat{g}_t) = \frac{\text{Cov}_P(A)}{n_P} + \frac{(1 - \alpha)^2 \text{Cov}_G(B)}{n_G}. \quad (9)$$

This expression includes covariance between direct and residual terms inside A . It gives no universal advantage over on-policy sampling; the frozen-policy audit chooses α from a preregistered grid using projected MSE times measured unit cost, then freezes it for confirmation. Nuisance fitting, α selection, and confirmation use three disjoint prompt partitions; the projection seed, dimension, reference sample, and confirmation replications are frozen before selection. Before these partitions are formed, every prompt is tied to a pinned source revision, export-byte hash, stable upstream identifier, raw-row hash, and versioned normalized-content hash. The downstream partitioner assigns every retained duplicate cluster wholly to fit, selection, or confirmation;

any later admitted hidden evaluator is a separate registered role. Semantic overlap with such evaluators is a blocking audit rather than being inferred from normalized-hash inequality. A hash-bound preflight rejects named PII/secret patterns and retained character-5-gram near-duplicates, but this lexical heuristic does not replace reviewed multilingual, paraphrase, or hidden-evaluator leakage analysis.

h_t^R, h_t^C, b_t , and q_t are trained only from data available before iteration t , then frozen. The sums below range over the complete finite prompt multiset, and prompts are sampled with replacement. A convenient adaptive proposal is

$$a_t(x) = d(x) \sqrt{\hat{v}_t(x)/\hat{c}_t(x)}, \quad (10)$$

$$Z_t^q = \sum_{x'} a_t(x'), \quad (11)$$

$$\bar{q}_t(x) = a_t(x)/Z_t^q, \quad (12)$$

$$q_t(x) = \epsilon d(x) + (1 - \epsilon) \bar{q}_t(x). \quad (13)$$

where $\hat{c}_t(x) > 0$ predicts generation cost and $\hat{v}_t(x)$ uses prior-iteration random projections to predict stale-source residual-gradient second moment. Because q also changes the mixture weight and its covariance with the direct term, this square-root rule is a heuristic, not an optimum for Equation (8). Incorrect predictors may reduce efficiency but do not alter the density-route expectation when the logged distribution and ratio are exact. We impose $\hat{v}_t(x) \geq v_{\min} > 0$ before normalization; equivalently, an implementation may set $\bar{q}_t = D$ whenever $Z_t^q = 0$.

4.2. Feedback and Credit Assignment

The main study uses deterministic programmatic verifiers: normalized answer checking for mathematics and isolated executable tests for code. Parser failures, timeouts, sandbox errors, and truncations are separate logged outcomes with predeclared scalar rewards rather than silently dropped samples. Hidden tests are disjoint from the authors' SFT, nuisance-model, and prompt-allocation data; unknown base-model pretraining contamination remains a limitation. Source and prompt draws are assigned before distributed generation. A valid sequence that reaches any declared verifier status is retained regardless of completion time. An infrastructure failure before a scored sequence exists aborts the complete batch, charges incurred cost, and performs no update; workers are never refilled selectively. This prevents selective refill but does not make completion ignorable. The theorem-facing audit therefore requires every preassigned draw to reach a scored outcome almost surely. If any infrastructure failure occurs, that replication is retained as an operational failure, never retried, and excluded from the estimator claim; H1 cannot pass from a success-conditioned subset.

Terminal U_λ multiplies the unnormalized sequence score in

Algorithm 1 One Adaptive-Mixture Update

- 1: Freeze $\theta_t, \lambda_t, h_t^R, h_t^C, b_t, q_t, \mu_t$, and α .
- 2: Draw n_P complete trajectories from $D\pi_{\theta_t}$.
- 3: Draw n_G new, single-use trajectories from $Q_t\mu_t$.
- 4: Log exact prompt, source, decoder, and sequence probabilities.
- 5: Evaluate verifier reward and token cost; form $U = R - \lambda_t C$.
- 6: Compute detached $w = P_{\theta_t}/M_t$ and Equation (8).
- 7: Finish accumulation at θ_t ; call `optimizer.step()` once.
- 8: Consume every trajectory; no microbatch update or second epoch.
- 9: Update λ from fresh current-policy token costs.
- 10: Add consumed data only to future nuisance-model training.

Equation (4); the theorem does not assign token-level causal credit. Reward shaping, group centering, group standardization, token-average losses, ratio clipping, and gradient clipping all change either the estimand or estimator and are excluded from the theorem-facing condition. They may be evaluated as clearly labeled practical variants. Shuffled, random, weakly related, and anti-correlated verifier controls test whether improvement follows the intended signal rather than pretrained behavior or evaluation leakage.

4.3. Algorithm and Complexity

Relative to on-policy REINFORCE, the method adds stale-actor generation, current and stale sequence log-probabilities under both policies, and nuisance-model inference. The cost ledger includes all generated tokens, verifier calls, likelihood evaluations, nuisance training, and policy forwards/backwards. Storage is linear in the single-use queue. A practical repeated-replay variant can reduce roll-out cost, but is outside the theorem and is reported as biased.

5. Analysis

Theorem 5.1. *Condition on $\mathcal{F}_t$. Assume (i) all policies, prompt laws, nuisance models, counts, and α are fixed before sampling; (ii) Equation (1) is differentiable and differentiation and expectation may be interchanged; (iii) the deterministic reward does not directly depend on θ ; (iv) b_t depends only on X ; (v) current and stale samples are conditionally independent, newly drawn, and have not influenced θ_t ; (vi) P_t and G_t have exactly evaluable masses on a common sample space, so M_t is computed exactly; (vii) every preassigned draw reaches a scored outcome almost surely; and (viii) required moments are finite. With exact Equation (7) and none of the clipping or normalization*

operations excluded above,

$$\mathbb{E}[\hat{g}_t \mid \mathcal{F}_t] = \nabla_{\theta} \mathcal{L}(\theta_t, \lambda_t). \quad (14)$$

The proof expands the two strata:

$$\begin{aligned} \mathbb{E}[\hat{g}_t] &= \mathbb{E}_P[(h - b)s] + \alpha \mathbb{E}_P[w(U - h)s] \\ &\quad + (1 - \alpha) \mathbb{E}_G[w(U - h)s] \end{aligned} \quad (15)$$

$$\begin{aligned} &= \mathbb{E}_P[(h - b)s] + \mathbb{E}_M[w(U - h)s] \\ &= \mathbb{E}_P[(U - b)s] = \mathbb{E}_P[Us]. \end{aligned} \quad (16)$$

The last equality is the prompt-baseline identity.

If $\tilde{w}(z)$ is the same fixed, pointwise implemented weight in both strata and $m_0(z) = \mathbb{E}[U \mid Z = z]$, the population bias is

$$\mathbb{E}_M[(\tilde{w} - w)(m_0 - h)s]. \quad (17)$$

This is a weight-misspecification identity only under frozen common weights. For source-specific fixed implementations it becomes

$$\alpha \mathbb{E}_P[(\tilde{w}_P - w)(m_0 - h)s] + (1 - \alpha) \mathbb{E}_G[(\tilde{w}_G - w)(m_0 - h)s]. \quad (18)$$

Thus either exact density ratios or an exact action-conditional outcome model removes this bias. Low prediction error or calibration alone is insufficient, and focal-batch nuisance fitting, batch-normalized weights, and data-dependent weights are not covered. In deterministic RLVR, $m_0(z) = U(z)$, so an exact outcome model makes the stale residual vanish and leaves an on-policy direct estimator. We therefore describe Equation (8) as augmented importance sampling rather than relying on a broad “doubly robust replay” label.

Scope of the result. Theorem 5.1 applies before one update. After the policy changes, the current component of M_t is no longer the target policy, the $1/\alpha$ bound need not hold, and the same data influenced the updated policy. Multiple epochs, ratio recomputation, or small KL do not restore the theorem. Optimizer momentum and clipping may still be useful, but the theorem is about the raw estimator rather than the transformed displacement or monotonic improvement.

6. Experiments

Evaluation starts with a deterministic software Gate 0 on a finite trajectory law. It must exactly reproduce the target gradient, mixture normalization, weight bound, and analytic bias of each injected ratio error before Phase A. Phase A then establishes estimator behavior at a frozen language-model policy before Phase B makes training claims. The primary hypotheses are: **H1**, exact adaptive-mixture augmentation reduces projected gradient MSE relative to on-policy estimation at matched total compute; **H2**, any reduction translates to better held-out reward per generated token

under one-update-per-batch training; and **H3**, the expected-response-length constraint produces a better reward-length frontier than implicit response-length normalization. Each can fail independently.

6.1. Experimental Setup

Frozen-policy audit. For each of two model families and both math and code verifiers, freeze a current policy, one stale policy, nuisance models, prompts, and decoders. For each target law generate two mutually independent large P -only reference estimates R_1, R_2 , independent also of candidate estimate $\hat{g}$, on adapter layers and projections frozen before outcomes. We estimate projected MSE by

$$\widehat{\text{MSE}}_{\times} = (\hat{g} - R_1)^{\top} (\hat{g} - R_2). \quad (19)$$

Its expectation is projected MSE although one realization may be negative. Intervals independently resample candidate replications, R_1 trajectories, and R_2 trajectories, retaining their shared-reference dependence. Selection and confirmation use separate reference pairs under their respective laws. Nuisance and proposal models are trained on disjoint data. Repeated subsamples compare on-policy policy gradient, ordinary importance sampling, bounded mixture importance sampling, augmentation with several predictor qualities, and adaptive versus uniform Q_t . Endpoints are bias, variance, MSE, interval coverage, effective sample size (ESS), and compute. Fault injection deliberately omits prompt ratios, uses base-model rather than decoder probabilities, fits the focal nuisance outcome, clips weights, and reuses data that formed the target policy. The committed finite-law Gate 0 currently executes the omitted-prompt-ratio and clipping cases. Base-versus-decoder scoring, focal nuisance fitting, and reuse are pending executable validators and must pass before Phase A; listing them here is a test plan, not evidence that they have run. The committed public-data configuration is a one-family mathematics pre-pilot for H1 only. It uses disjoint normalized laws D_{fit} , D_{select} , and D_{confirm} ; selection MSE is not described as an estimate under the confirmation or full-frame law. For $\alpha \in \{.25, .5, .75, 1\}$ it fixes $(n_P, n_G) \in \{(16, 48), (32, 32), (48, 16), (64, 0)\}$ and minimizes projected MSE times measured accelerator-seconds at a common 64-trajectory and verifier-call budget. Tying mixture mass to allocation fraction is a pilot restriction, not a condition for unbiasedness; a confirmatory design must vary them separately. Code, second-family, H2, and H3 confirmatory runs require a later frozen configuration.

End-to-end training. Confirmatory training uses two open model families at small and medium scale, one exact-answer math suite and one sandboxed code suite. Prompts, hidden verifiers, maximum generation length, temperature, optimizer, and checkpoint selection rules are frozen before

the main runs. Each collected batch produces one update and is never used again by the theorem-facing method. Staleness is stratified by measured policy KL, not checkpoint age.

Baselines are on-policy RLOO, Dr. GRPO, separately tuned DAPO and GSPO, TOPR, variance-informed prompt allocation (VIP), DBB reward estimation, PrefixRL, PROS, SPEC-RL, MinPRO, the ICML replay design, POPO, plain bounded-mixture importance sampling, augmented sampling with uniform Q , and the full adaptive proposal. All generated prefixes and complete sequences, verifier calls, likelihood passes, nuisance training, policy updates, and hyperparameter searches are charged. Each configuration receives the same token and verifier-call budget, while total allocated accelerator and CPU time is reported as a Pareto cost rather than assumed equal. We use five independent development seeds and five frozen confirmatory seeds. Source-isolation controls set $\mu_t = \pi_t$ while adapting Q_t , and set $Q_t = D$ while varying policy staleness. Rollout counts are fixed before each batch. A batch is admitted only if its pre-declared worst-case token and verifier-call reserve fits; the run stops at the last fully admitted batch and reports budget undershoot. Realized token, call, accelerator, and CPU costs are never converted into one unspecified scalar. Reward-versus-realized-token and reward-versus-accelerator curves are primary; endpoint contrasts are reported only at their largest common realized-cost checkpoint, with no interpolation across updates.

6.2. Main Results

Every value in Table 1 will aggregate independent runs with a predeclared interval and link to immutable trajectory manifests. The repeated-replay row is explicitly not covered by Theorem 5.1; it tests whether a biased engineering variant is nevertheless more useful.

6.3. Ablations and Diagnostics

We ablate the on-policy anchor fraction α , augmentation, adaptive prompt proposal, prompt baseline, cross-iteration nuisance freezing, and length dual. A separate deliberate-violation study runs 1, 2, 4, and 8 epochs; uses fresh, previously consumed, and reward-selected stale data; and compares unclipped and clipped ratios. After each update, a large fresh on-policy batch estimates the resulting approximation error.

Required diagnostics include the numerical $1/\alpha$ bound; mixture normalization; joint and source-specific ESS; maximum weight and support failures; direct and residual gradient norms and cosine; outcome calibration by length, difficulty, and source; fresh versus augmented projected gradients; pre/post-update KL; entropy; sign-conditioned update mass; policy-to-behavior KL; dual trajectory; reward-token fron-

tier; parser errors; and verifier disagreement. Reporting only reward curves is insufficient to test the method. The length study freezes $\kappa \in \{512, 1024, 2048\}$, $\lambda_0 = 0$, and dual step size 10^{-4} ; last-iterate and final-20%-average frontiers are both reported. Replay comparators freeze admission, buffer capacity, reuse count, KL stratum, clipping, and update count, with original generation and rescoring charged.

6.4. Generalization and Safety

Generalization holds the estimator configuration fixed across model family, scale, prompt difficulty, context length, and verifier domain. Hidden verifier variants alter answer formatting, unit tests, parser implementation, and timeouts. Reward controls include shuffled labels, random rewards, weak heuristics, and anti-correlated signals. We measure train/hidden reward gaps, pass@1 and pass@ k , response-length quantiles, invalid-output rates, memorized-answer indicators, and diversity. Safety claims are not inferred from verifier reward; the protocol will report separate capability and harmful-behavior audits as possible regressions.

7. Limitations

The assumptions intentionally exclude the main appeal of replay: repeated use of old data. Single-use stale actors may add systems complexity without enough variance reduction to justify it. Exact sequence probabilities are expensive and fragile when production decoders use top- p , rejection, tool calls, or environment transitions. Adaptive prompt density must be logged exactly. Importance weights remain high-variance despite their bound, especially for small α and long horizons.

The outcome predictor is a critic by another name and may consume substantial compute. The two-route property in Equation (17) is practically asymmetric: behavior likelihoods are logged, whereas an exact sequence-conditional verifier predictor is unrealistic. Conditional gradient unbiasedness does not imply low MSE, stable Adam updates, monotonic improvement, or resistance to reward hacking. Finally, math and code evidence may not generalize to subjective human-feedback rewards or frontier models.

8. Conclusion

We separated statistically controlled stale generation from ordinary replay and specified an adaptive-mixture augmented policy gradient for expected-response-length-constrained RLVR. The estimator has a simple conditional expectation identity under strong, auditable restrictions, and fails to inherit that identity after data reuse or multiple updates. This clarity is useful only if the estimator actually reduces gradient error after all added compute is charged. The frozen-policy audit is therefore the next decisive step;

Table 1. Primary matched-budget results. Higher held-out reward is better; lower projected gradient MSE and generated tokens to threshold are better. Dashes mark measurements that have not yet been run.

Method	Gradient MSE	Held-out reward	Tokens to threshold	Mean response tokens
On-policy RLOO	–	–	–	–
Bounded mixture IS	–	–	–	–
Augmented mixture, uniform Q	–	–	–	–
Augmented mixture, adaptive Q	–	–	–	–
Practical repeated-replay variant	–	–	–	–

downstream training should proceed only if that test succeeds.

Impact Statement

More sample-efficient RL could reduce the compute required to improve reasoning systems, but it can equally accelerate optimization toward flawed or harmful verifiers. Exact-looking tests can encode incomplete specifications, reward insecure code, or favor brittle answer formats. Adaptive allocation may also neglect rare prompts unless its support floor is enforced. We therefore require hidden-verifier variants, random and adversarial reward controls, minimum prompt support, full token and energy accounting, and separate safety audits. Stale trajectories and prompts may contain sensitive content and must follow the same retention and access controls as training data. The paper does not equate verifiability with correctness, harmlessness, or social value.

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A. Algorithmic and Reproducibility Details

A.1. Dual Update

The expected-response-length dual uses only fresh target-policy trajectories:

$$\lambda_{t+1} = \left[\lambda_t + \eta_\lambda \left\{ \frac{1}{n_P} \sum_{i=1}^{n_P} C(Y_i^P) - \kappa \right\} \right]_+ . \quad (20)$$

κ is fixed before training. Policy and dual updates both estimate quantities at (θ_t, λ_t) and are committed after collection. Reward, response length, and violation are reported separately rather than only through U_λ . This projected stochastic controller does not guarantee that every iterate or the final policy is feasible. The constraint does not cover prompt tokens, likelihood passes, verifier cost, or total training compute; those belong to the experimental ledger.

A.2. Logging Contract

Each trajectory records prompt identifier and probability, source component, current and behavior checkpoint hashes, tokenizer, temperature, EOS and truncation rule, random seed, exact current and behavior sequence log-probabilities, reward components, parser status, generated tokens, verifier version, nuisance-model revision, use count, and policy-update identifier. The run aborts if a supposedly theorem-facing trajectory has use count above one or if its source probability cannot be reconstructed. For all candidate trajectories, including those in aborted batches, the ledger computes

$$K_{\text{gen}} = \sum_z C(z), \quad K_{\text{ver}} = \sum_z \mathbf{1}\{\text{verifier called on } z\}, \quad (21)$$

$$K_{\text{score}} = \sum_z C(z) N_{\text{added score}}(z). \quad (22)$$

No cost disappears when a trajectory is excluded from the estimator.

A.3. Protocol Manifest

The repository runner `python3 scripts/run_experiment_pipeline.py --plan experiments/rl_pipeline.json` enumerates immutable prompt acquisition, verifier processing, policy and nuisance-model construction, and a frozen-policy H1 estimator pre-pilot. The full end-to-end H2/H3 plan is not executable from this configuration. The committed JSON is a stage and artifact contract tied to the hashed RL section of `experiments/pilot_preregistration.json`; site-local commands are required to execute it, and each executed stage records output hashes and separate token, verifier, accelerator, CPU, and wall-time costs. The public source registry pins the prompt source and base checkpoint only. The current and stale policy tensor hashes, tokenizer/decoder hashes, KL sampling law, and mathematics-verifier revision are still unfrozen blockers; this H1 pre-pilot cannot execute until a replacement configuration records them. A hidden code evaluator is outside this mathematics-only pre-pilot and must be registered by any later code study. Automated license metadata is only a preflight: real acquisition remains blocked on manual provenance, terms, privacy, and leakage review. Once approved, `scripts/materialize_public_data.py` verifies each site-local export hash, emits row-level provenance, deduplicates within roles, and quarantines normalized-content collisions across roles. The committed fixture exercises those rules without downloading prompts; all of its outputs are marked non-empirical. `scripts/run_content_gates.py` also binds the frozen `experiments/content_gates.json` contract to the materialization hashes and fails closed on retained named PII/secret patterns or lexical near-duplicates. Because the large-artifact MinHash-LSH candidate screen can miss pairs, it is always inconclusive and blocking until a reviewed recall-complete audit is hash-bound. Even an exhaustive pass neither clears a source nor establishes semantic independence from a hidden evaluator. The companion `synthetic_commands.json` fixes a four-cell trajectory law, $\alpha = 1/2$, 32 draws per source, 2,000 replications, and seed 20260902. It enumerates the target and estimator expectations to machine precision, checks $\sum_z M(z)w(z) = 1$ and $w \leq 1/\alpha$, and verifies the bias identity for omitted-prompt-ratio and weight-clipping errors. Gate 0 fails closed if any identity differs by more than 10^{-12} . This is an equation and software audit, not model-training evidence; its outputs cannot populate Table 1.

A.4. Optional Prefix Gate

Prefix continuation is excluded from the core experiments. A later extension may draw a prefix $H \sim \mu_t$, choose a predictable $p_t(H) \in [p_{\min}, 1]$ before observing the suffix or reward, and include a completed stale-source residual with factor $I/p_t(H)$. The denominator remains the fixed number of candidate prefixes, not the number completed. Every discarded prefix is charged. Sequential gates require the product of all conditional continuation probabilities; group-relative estimators additionally require joint inclusion probabilities. This extension must be compared directly with PrefixRL and PROS (Setlur et al., 2026; Huang & Wan, 2026).

B. Additional Results and Failures

Table 2. Mandatory estimator-audit diagnostics; pending execution.

Diagnostic	Value
Projected bias relative to fresh target	–
Projected variance and MSE	–
Maximum mixture weight / $1/\alpha$	–
Joint effective sample size	–
Direct–residual gradient cosine	–
Prompt-density reconstruction error	–
Outcome calibration error	–
Added compute / on-policy compute	–

Complete learning curves, per-seed endpoints, theorem-violation results, reward-control trajectories, parser failures, and examples selected by a fixed random seed will be released whether they favor the method or not.